Richard Simmons

Full-Stack Engineer · AI · Product

Remote (US) · Canal Winchester, OH · 614-632-3839 · richard.simmons.dev@gmail.com
Portfolio (Three.js, custom GLSL shaders): richard-simmons-portfolio.vercel.app
github.com/RSimmons2021 · linkedin.com/in/richard-simmons-a3916958

STACK

Languages Python, TypeScript, JavaScript, SQL, Java
Frontend React, Next.js, React Native, TailwindCSS, Three.js / WebGL / GLSL, Vite
Backend FastAPI, Node.js, Flask, PostgreSQL, Redis, REST, background workers, microservices
AI LangGraph, LangChain, RAG, agentic workflows and tool calling, OpenAI / Anthropic APIs, structured outputs, evals and tracing, vector search
Infra AWS (S3, Lambda), Docker, GitHub Actions, Supabase, Vercel, serverless edge functions

EXPERIENCE

AI Engineer (Contract) June 2026 – Present
Loopp (FoundersMax AI Build Studio) · Remote

- Integrated AI agents into a legal firm's document and case-intake workflow, wiring LLM automation into systems already load-bearing for the business. In production with real users.
- Build client agent systems in Python, FastAPI, and LangGraph: multi-step reasoning, tool calling, and defined handoff criteria for when an agent should stop and escalate to a human.
- Raised document processing throughput by 500 documents per month, validated against an eval suite.

Founder & Lead Engineer April 2025 – June 2026
Zoan Collective LLC · Remote, Columbus, OH

- Built Lucid's first backend on self-hosted Meta MusicGen behind Flask, then threw it out. The output quality wasn't good enough to charge for and prompt tuning was never going to fix it. Rebuilt generation on Suno's API behind Supabase Edge Functions, deleted the server entirely, and cut infrastructure cost 80% and track delivery from 360s to 120s.
- Shipped two consumer apps (FocusFi, Lucid) to the App Store and Google Play solo: architecture, backend, UI, payments, and review process.
- Ran LLM and audio pipelines at 1,000+ interactions/day on background workers with request queuing and cache reuse, which is what actually killed the rate-limit failures. The concurrency was never the bottleneck; the vendor queue was.
- Handled StoreKit 2 and Google Play Billing end to end, including subscription edge cases: refunds, grace periods, and restores.

Software Engineer June 2021 – March 2025
Toyota · Remote

- Built and shipped REST services handling 1M+ daily requests on a manufacturing analytics platform, serving 10,000+ daily users across 7 North American facilities.
- Shipped 10+ production apps, from inventory systems through dealership sites, on React/TypeScript with Python/Flask and Node.js behind.
- Ran a UX audit across 3 enterprise modules used by 2,000+ plant staff and redesigned the navigation. Task completion time dropped 20%. The wins came from removing screens, not adding features.
- Took Jest coverage to 85% in CI; production bugs fell 15%. Reviewed frontend architecture and API design across time zones.

Junior Data Scientist June 2018 – August 2020
Knoesis Lab · Dayton, OH — classifiers on 50k+ social records (Pandas, NumPy), 87% accuracy, published research

SELECTED PROJECTS

Agent Lab (LLM Planner Eval Harness) ai-agent-portal-site.vercel.app
Three.js · Cannon.js · OpenAI / Anthropic APIs · Node.js · Vite

- Evaluation harness for LLM planners on a working Portal-style 3D game. The browser sends a structured observation, not pixels, to a planner endpoint with one strict tool schema across the OpenAI and Anthropic APIs. The physics engine stays authoritative: a planner that claims success without reaching the goal scores 0/24.
- Headless eval suite reports verified success rate, invalid tool-call rate, p50/p95 latency, and cost per verified episode by puzzle kind. Attaching per-candidate attempt history to the observation took the small model from 2/24 to 20/24 on the hardest tier and cut cost per verified episode 12x, with no model or budget change.

Agent Relay (Agent Permission Control Center) openai-agent-site.vercel.app
React / Vite · FastAPI · LangGraph · SQLite · SSE · Ed25519 / WebCrypto

- Agent permission layer where authority is deterministic, not model-decided. A LangGraph pipeline resolves identity, fetches each service's capability manifest over HTTP, and returns exactly one of AUTO_EXECUTE, APPROVAL_REQUIRED, or BLOCKED. The model only explains that result and never becomes authority. Approving re-runs the graph against a freshly fetched manifest instead of resuming in place.
- Every control action is Ed25519-signed over method, path, and body hash, so a captured signature cannot be replayed or edited from deny to approve. The adversarial suite feeds paraphrased injections the substring tripwire misses and requires the contract to block all of them with the tripwire silent. CI gates that containment at 100% across 46 unit tests and 35 eval scenarios.

EDUCATION

B.S. Computer Science May 2021
Wright State University · Dayton, OH

OUTSIDE THE JOB DESCRIPTION

- Music producer, 10+ years. Why I could hear MusicGen wasn't good enough before any metric told me. Design obsessive: Dieter Rams, Zaha Hadid, Grasshopper. Home lab: self-hosted, local-first.